COOLSHADE · DESIGN DECISION RECORD

CoolShade concept options

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CSH-DDR-004	Version	0.1
Status	Draft	Date	2026-10-30
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coolshade		
Source	docs/decisions/0004-concept-options.md @ ae091ec		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-30	Amish Chadha	Concept options compared in the kit 1.9.1 upgrade (CSH-OPT-001); the choice is proposed, awaiting Amish

0004: Concept options

- **Date:** 2026-10-30
- **Status:** proposed, awaiting Amish. Nothing in this record is decided.

Context

Kit 1.8 asks every design to compare at least two working principles before it commits (STANDARDS section 21). CoolShade was designed at kit 1.7 as a solar misting shade canopy; in this record that design is option A and it does not change. Two further options with different working principles were added and all three were modelled to detail level D3 ("Detail every option", Amish, 2026-10-24). The options, their numbers and the scoring are in CSH-OPT-001; their STEM screens are in CSH-STM-001 (S22 to S26).

Options considered

1. **Option A, solar misting shade canopy.** Score 134 of 190. The current design: 7 bar mist from eight anti-drip nozzles, switched on heat, humidity and presence; 2.7 °C cooling while spraying for a group. Open: wetting (S4), *Legionella* (S14), cost USD 882 against USD 775.
2. **Option B, wet-pad fan cooler under shade.** Score 135. Direct evaporative cooling through a cellulose pad and a 12 V fan, once-through water, no free aerosol. It reaches one or two people (about 1.5 kW), has a welded cabinet and uses about 2.2 L/h.
3. **Option C, passive double-skin reflective roof.** Score 159. A white composite skin 100 mm above the cloth; no water, no power, no moving parts. It cools no air (UR-002 scores 1), so it is a better shade, not a cooling product, and the skin cannot come off before a gust.

Decision

Proposed, awaiting Amish. Options A and B are within 10 % of each other, so both go to Amish. Option C scores highest, but it fails a High need, so the score does not make it the pick.

Recommendation: keep **option A** as the design, consider **option C's white skin as an add-on to A** (to assess; not modelled or priced), and keep **option B as the fallback** if the droplets are too big or the hygiene plan fails. `concept_chosen` is "A (proposed)" in `project.yaml`.

Consequences

- If Amish confirms A: no design change; the precis, requirements, calculation note, drawing, BOM and build plan already describe A.
- If Amish chooses B or C: the PRS, CAL, BOM, drawing and build plan would be redone for that option; their D3 models exist (`cad/src/csh_options.py`) but have no priced BOM, drawing or build plan.
- If Amish asks for the hybrid: the roof would carry both the removable cloth and a fixed skin, and the wind case (cloth off above 15 m/s, S17) would be redone.